

Hyperloop Development Program

Reporting Period 7: 1 Jan 2024 – 30 June 2024

24-09-2024

V1.0

This report highlights the progress of the Hyperloop Development Program. First section summarizes the highlights within the growing hyperloop market, the second section provides an overall program overview, followed by section of progress per working group, and summarized in the financial overview.

The Hyperloop Development Program has requested a renewal extension, showing progress despite funding delays and permitting challenges, with the construction of the EHC underway and new partnerships established. The plan has been adjusted, incorporating changes in Working Groups deliverables, budgets, and time estimations.

REPORTING ON HIGHLIGHTS WITHIN HYPERLOOP MARKET

The past months have shown highlights in the Global developments in the Hyperloop Market:

January

- Webuild, Leonardo and partners win the European tender for a hyperloop route – the Hyper Transfer project – in Veneto, Italy.
- Europe's Rail Joint Undertaking (EU-Rail) launches a call for proposals, which is aimed at the technical harmonization, the design concept, and roadmaps on industrialization for the hyperloop.
- ArcelorMittal pairs up with Indian partners to develop hyperloop.

February

- The China Aerospace Science and Industry Corporation successfully performs a test with hyperloop-related technology, breaking a speed record of more than 600 km/h on a 2 km test track in Shanxi, China.
- The European Hyperloop Center and the EuroTube Foundation decide to closely work together within a European hyperloop infrastructure cluster that is facilitated by the Hyperloop Development Program.
- Klaus Rudischhauser, former Ambassador of the EU to Mexico, joins the HDP as Director General.

March

- The European Hyperloop Center draws global news coverage after unveiling its infrastructure that is fully equipped for testing and demonstrating.
- Indian Railways launches a tender for a demonstrative hyperloop project.

April

- The Institute of Hyperloop Technology starts the construction of the goTube hyperloop facility for which they acquired magnetic propulsion technology and power electronics from Polish hyperloop developer Nevomo.

- Zeleros, the Spanish hyperloop developer, attracts new investment and embarks on a new stage by focusing on the development of batteries, electrification technologies, and transportation automation systems.
- The Indian Institute of Technology in Madras announces a hyperloop competition for 2025.

May

- In China concepts are being developed for a very first hyperloop corridor by 2035, connecting Shanghai with Hangzhou over a distance of 150 km.
- EuroTube starts the assembly of its hyperloop test facility, spanning 120 meters, in Dübendorf, Switzerland.
- Swisspod finishes the first construction phase of their hyperloop test track in Colorado, USA.

June

- The EU recognizes hyperloop as an opportunity for more sustainable technology in its transport network policy (TEN-T).
- The German Railway Authority (Eisenbahn-Bundesamt) initiates a research project for assessing the current state of hyperloop development. The project aims to provide objective results, including information gaps.
- Student team Delft Hyperloop demonstrates their lane switch technology at the Technological University of Delft.

REPORTING ON OVERALL PROGRAM PROGRESS

This report is to confirm costs incurred and progress in the Hyperloop Development Program over the period 1 Jan 2024 – 30 June 2024.

The key program deliverable and key deliverable for this stage per working group are represented below.

WG	Key Program deliverable	Key deliverable Status
WG Hyperloop Development	Demonstrated hyperloop for cargo system (TRL-7) in the EHC	Vehicle commissioned and at EHC for the first tests.
WG EHC	European Hyperloop Center (EHC) Phase A completed	EHC infrastructure has been delivered, started with facilitating the first vehicle tests.
WG FP	Identify and develop 5 routes up to a preliminary level, of which 1 with a go ahead build-out of route	Research Infrastructure Cluster formed, working towards the Test & Demonstration goals of a future testing facility.
WG Program Management	Establish a connected hyperloop ecosystem	Follow-up project initiated with Hyper4Rail in which HDP is the coordinator

HDP is continuing Stage 3 of a total of 4 stages – as there have been encountered delays, the stage 3 is extended and concluded in June 2024. The Program will wrap up by the end of 2024, with the final stage will be concluded over 2024-H2.

The progress of the Working Groups and Work Packages is presented in figure 1. Green means the Work Package is delivered; yellow the Work Package is on-going.

PLANNING FOR PHASE 2020-2024

STATUS: AUG 2024

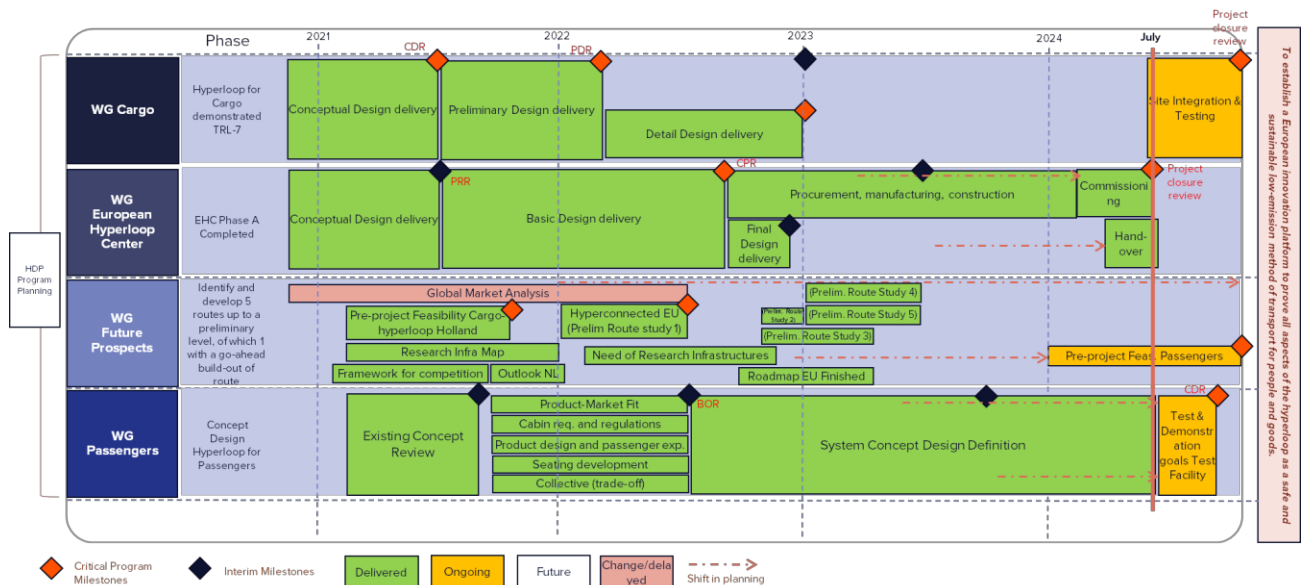


Figure 1: Stages & Milestone planning HDP

The detailed overview of program deliverable status for open stage 1, stage 2, stage 3, and stage 4 is presented below. A number of milestones from stage 1, 2 and 3 have encountered delays, all adjusted delivery dates are presented below.

#	Milestone	Working Group	Status	(Adj.) Date delivery
Open from Stage 1:				
1.9	Final Market opportunity attractiveness	Future Prospects	Ongoing	Q4 2024
Open from Stage 2:				
2.8	Scheme of operations for Hyperloop Research Infrastructure	Future Prospects	On-going	Q2 2024
Stage 3:				
3.9	EHC Delivery & Start first tests	EHC	On-going	Q3 2024
3.10	Project hand-over EHC	EHC	Finished	Q1 2024
3.12	Work Package Research Infra finished	Future Prospects	On-going	Q4 2024
Stage 4:				
4.1	Hyperloop Demonstrated at EHC	EHC	On-going	Q3 2024
4.6	Test & Demonstration goals Passenger Test facility	Future Prospects	On-going	Q4 2024

REPORTING PER WORKING GROUP

WG Hyperloop Development

Status: Proceeding according to WG Plan

Within the WG Cargo major progress has been made with respect to the manufacturing, assembly and delivery of the European Hyperloop Center. Highlights include:

Vehicle Development

The following is a summary in images:

- The vehicle assembly has been completed, and the commissioning phase is underway.
- Initial tests of levitation, guidance, and propulsion have been successfully conducted.
- A development test rig has been established in Rotterdam, where the first levitation, guidance, and propulsion tests took place. The vehicle is scheduled for transport to the European Hyperloop Center in July/August.



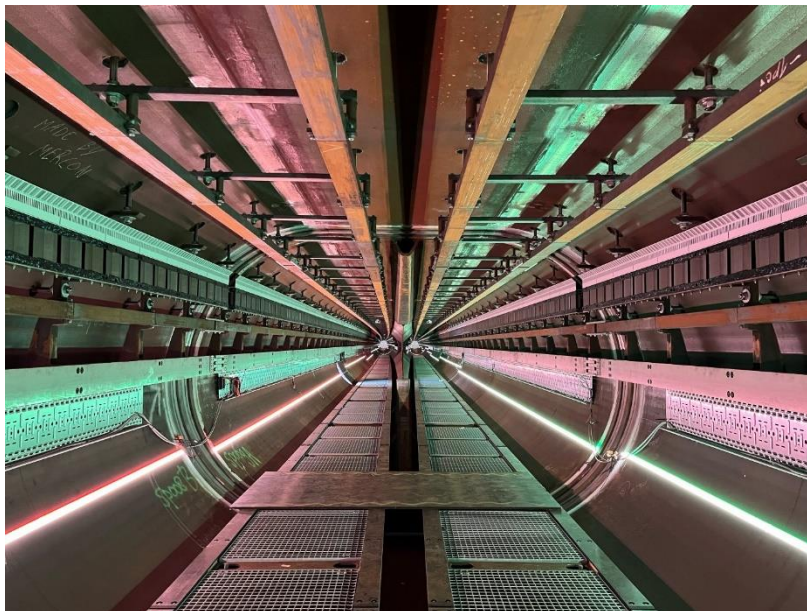
WG EHC

Status: The project is on track to complete the test facility by March 31st, 2024, with the official handover planned for Q3.

Commissioning Progress:

- Significant progress has been made in adjusting the on-site infrastructure and related activities.
- A workshop has been constructed adjacent to the main infrastructure.
- Electrical infrastructure has advanced, with cameras installed, internal lighting connected, and sensors in place.
- Cabinets have been installed at Veendam.

Handover: The facility handover is scheduled for completion by March 31st, 2024. This will be followed by a "Test After Completion" phase, during which the remaining punch list for the infrastructure will be addressed, with completion expected by September 2024.



EHC Vision for next phase

We had the privilege of welcoming key leaders from the Province of Groningen and the Municipality of Veendam. Our team provided an in-depth presentation on the latest hyperloop developments and highlighted the unique opportunities offered by the European Hyperloop Center (EHC).

A significant milestone was achieved as we shared the EHC's Vision Document with Deputy Directors Bram Schmaal (Internationalization) and Henk Emmens (Economics) from the Province of Groningen, alongside Alderman Henk Jan Schmaal of Veendam.

Activity report

EuroTube H1 2024

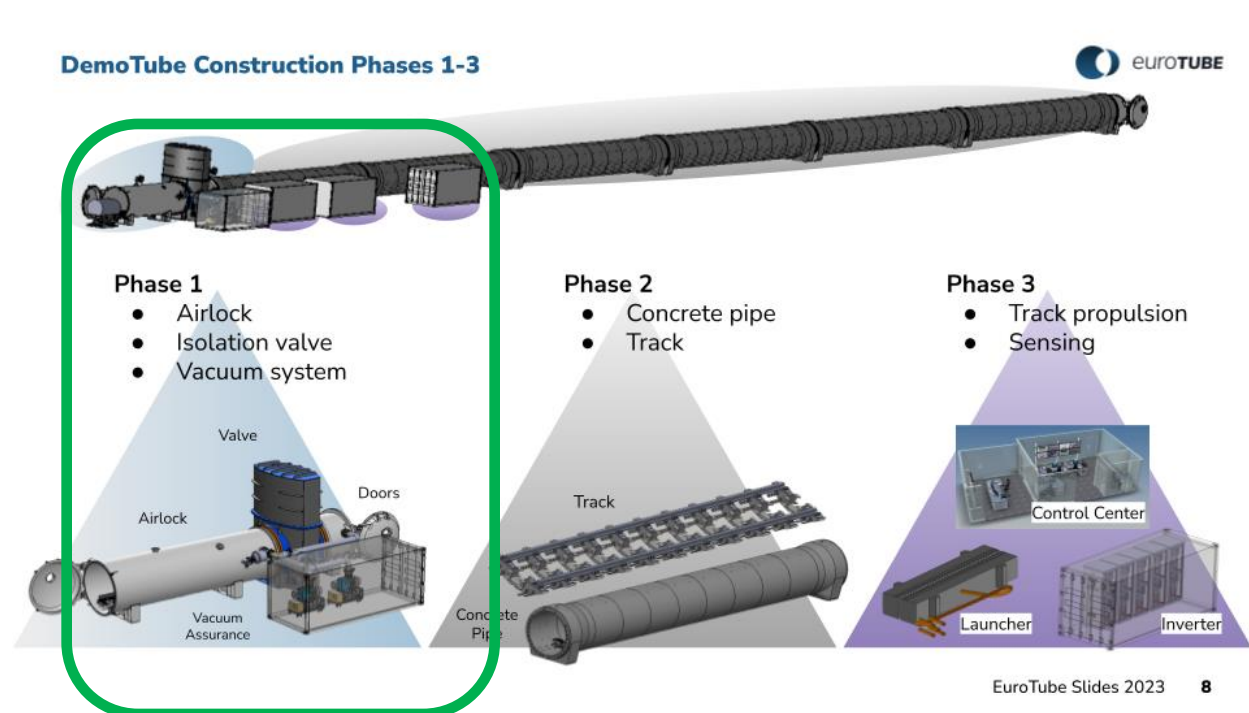
1 WG EHC - DemoTube

In Q2 2024, EuroTube has kicked-off the construction of *DemoTube*, which is a 120m long demonstrator. As described further below, it's main purpose is the proof of concept of construction processes and the concept of sealed, modular post-tensioned concrete tubes. Since all construction steps are performed by EuroTube

itself to maximize learning and measurement and data gain, construction will last until mid-2025.

In June, in time for the European Hyperloop Week (EHW), the Phase 1 of DemoTube was completed. This meant the assembly of the airlock, connecting it to the BigValve and assuring its vacuum-tightness. Phase 1 also included the interior refurbishment of the Vacuum and control containers to fit the needs, piping, electrical and pneumatic works. Finally, the Control System was developed. With that, the main „active parts“ of DemoTube are completed and vacuum test within the airlock can be run.

The BigValve, the first valve suited for hyperloop application, was successfully tested. During EHW, the pod of Swissloop was tested in DemoTube for its leakage compensation mechanism.





WG Future Prospects

Status: Ongoing

The following activities are summarized:

The HDP testing infrastructure cluster

In order to maximize the knowledge gained from these infrastructures, HDP has, together with the European hyperloop center (EHC) and EuroTube, started the "HDP testing infrastructure cluster", to align European Test facilities into a collaborative cluster. As part of this Infrastructure Cluster, the partners have been working on a Test & Demonstration plan for passenger test facility, reported under "Future hyperloop testing infrastructure".

Building International Relationships for Ecosystem Expansion

HDP has been initiated the application for a large hyperloop concept harmonization project initiated by the European Commission under Europe's Rail Joint Undertaking (ERJU): AWP24-FA7: ROADMAP TOWARDS INDUSTRIALISATION AND HARMONIZED IMPLEMENTABLE CONCEPT. The bid was handed in in May. Decision expected in Q3-'24.